

1. Administrative & Study Identification

Full Study Title: A Phase III Double-blind Randomised Study Assessing the Efficacy and Safety of Capivasertib + Fulvestrant Versus Placebo + Fulvestrant as Treatment for Locally Advanced (Inoperable) or Metastatic Hormone Receptor Positive, Human Epidermal Growth Factor Receptor 2 Negative (HR+/HER2-) Breast Cancer Following Recurrence or Progression On or After Treatment With an Aromatase Inhibitor **Short Title/Acronym:** CAPItello-291 **Protocol Number:** D3615C00001 **NCT Number:** NCT04305496 **Trial Status:** ACTIVE_NOT_RECRUITING **Sponsor/Company:** AstraZeneca **Investigational Product:** Capivasertib (Truqap) + Fulvestrant (Trade Name: Faslodex | ATC Code: L02BA03) **Phase of Development:** Phase III **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of capivasertib in combination with fulvestrant versus placebo with fulvestrant by assessing investigator-assessed Progression-Free Survival (PFS) in the overall population and in patients whose tumors have PIK3CA/AKT1/PTEN-alterations. **Secondary Objectives:** To evaluate Overall Survival (OS), Objective Response Rate (ORR), Duration of Response (DoR), patient-reported health-related quality of life (HRQOL), and overall safety and tolerability.

2.2 Background & Rationale To address the need for effective targeted treatments in adult patients with hormone receptor (HR)-positive, human epidermal growth factor receptor 2 (HER2)-negative locally advanced or metastatic breast cancer. Specifically, this study was designed for patients who have progressed on aromatase inhibitor-based therapy or recurred within 12 months of completing adjuvant therapy, leveraging capivasertib to block AKT protein activity and overcome resistance to standard endocrine therapies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1

3.2 Planned Enrollment & Duration Targeted Enrollment: 818 patients **Number of Sites:** Multicenter (Global) **Estimated Study Start:** June 2020 **Estimated Primary Completion:** June 2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult females (pre- and/or post-menopausal) and adult males with histologically confirmed HR+/HER2- breast cancer. Pre-menopausal (and peri-menopausal) women can be enrolled if amenable to treatment with an LHRH agonist (must commence concomitant treatment prior to or on Cycle 1, Day 1 and continue for the duration of the study). 2. Metastatic or locally advanced disease not amenable to resection with curative intent, with radiological or objective evidence of recurrence or progression. 3. Disease recurrence or progression while on, or within 12 months of completing, an aromatase inhibitor (AI)-containing regimen, or radiological progression on prior AI administered for advanced disease. 4. ECOG/WHO Performance Status: 0-1. 5. Patients must have measurable disease according to RECIST 1.1 and/or at least 1 lytic or mixed (lytic + sclerotic) bone lesion that can be assessed by CT or MRI. 6. Provision of a Formalin-Fixed Paraffin-Embedded (FFPE) tumour sample from primary/recurrent cancer for central testing. 7. Allowed to have received up to two prior lines of endocrine therapy and up to 1 line of chemotherapy for locally advanced or metastatic disease.

4.2 Exclusion Criteria 1. Symptomatic visceral disease or any disease burden that makes the patient ineligible for endocrine therapy. 2. Prior treatment with AKT, PI3K, or mTOR inhibitors, Fulvestrant, or other SERDs. 3. Treatment with other chemotherapy, immunotherapy, immunosuppressant medication, or anticancer agents within 3 weeks prior, or potent inhibitors/inducers of CYP3A4 within 2 weeks prior to study treatment initiation. 4. Radiotherapy with a wide field of radiation up to 4 weeks, or limited field for palliation up to 2 weeks, before study treatment initiation. 5. Any unresolved toxicities from prior therapy greater than CTCAE grade 1 at the time of starting study treatment (with the exception of alopecia). 6. Spinal cord compression or brain metastases unless asymptomatic, treated, stable, and not requiring steroids up to 4 weeks before study treatment initiation. 7. Any clinically important cardiac abnormalities (e.g., QTc >470 msec, left bundle branch block, uncontrolled hypotension <90/50 mmHg, EF <50%, or recent history of myocardial infarction/CABG/angina/CHF NYHA grade ≥2). 8. Clinically significant abnormalities of glucose metabolism (e.g., Type 1 or 2 diabetes requiring insulin treatment, or HbA1c ≥8.0%). 9. Known abnormalities in coagulation (e.g., bleeding diathesis) or treatment with anticoagulants precluding intramuscular injections of fulvestrant or LHRH agonist. 10. Currently pregnant (confirmed with positive pregnancy test) or breast-feeding.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Capivasertib 400 mg (2 oral tablets) twice daily (approximately 12 hours apart) for 4

days followed by 3 days off each week over a 28-day treatment cycle. Fulvestrant 500 mg on Cycle 1 Days 1 and 15, and then Day 1 of each 28-day cycle thereafter. **Route of Administration:** Capivasertib is administered orally; Fulvestrant is administered intramuscularly. **Duration of Treatment:** Continued until disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care measures, concomitant LHRH agonists for pre/perimenopausal women. **Prohibited:** Concurrent systemic anticancer therapy; strong inhibitors/inducers of CYP3A4, and medications known to prolong the QT interval.

6. Study Timeline & Visit Schedule

Screening: Day -28 to 0 (Informed Consent, PIK3CA/AKT1/PTEN testing via companion diagnostic, baseline scans/ECG). **Baseline:** Day 0 (Randomization, First dose administration). **Interim:** Cycle 1 Days 1 and 15 (Fulvestrant loading), then Day 1 of every 28-day cycle (Safety Labs, Efficacy Assessments, Patient-Reported Outcomes, Medication compliance for intermittent 4-days-on/3-days-off oral regimen). **End of Study:** At disease progression or unacceptable toxicity (Final Vital Signs, Exit Interview, Follow-up for overall survival).

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression Free Survival (Months and Percentage): Defined as the time from randomization until progression per RECIST v1.1, as assessed by the investigator at the local site, or death due to any cause. Assessed in both the Global Cohort Overall Population and the Altered Population. **Measurement Method:** Tumor assessments evaluated according to Response Evaluation Criteria in Solid Tumors (RECIST), version 1.1 (at least a 20% increase in the sum of the longest diameter of target lesions, clinically significant increase in a non-target lesion, or appearance of new lesions).

7.2 Secondary & Exploratory Endpoints 1. Overall Survival (OS) 2. Objective Response Rate (ORR) and Duration of Response (DoR) 3. Health-related quality of life (HRQOL) assessed via EORTC QLQ-C30 and PRO-CTCAE responses

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine vital signs, ECG tracking, lab assessments (especially monitoring for glucose metabolism and cutaneous adverse reactions/rash), and patient-reported PRO-CTCAE diaries. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline for severe/life-threatening events. **Data Monitoring Committee (DMC):** External independent committee to evaluate ongoing safety, especially specific adverse events such as severe diarrhea, hyperglycemia, and stomatitis.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) Population for primary efficacy analysis; Safety Set for AE/SAE analyses. **Sample Size Justification:** Powered to detect statistically significant improvements in PFS in both the general enrolled population and the biomarker-positive subpopulation (PIK3CA/AKT1/PTEN altered). **Handling of Missing Data:** Time-to-event data analyzed utilizing the Kaplan-Meier estimate, with appropriate censoring mechanisms for patients lacking progression documentation. (Participants discontinuing treatment prior to progression continue to be scanned until progression).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite and remote clinical monitoring by the sponsor to ensure protocol adherence and safety. **Audit Trail:** Formal eCRF locking processes and Independent Review Facility (IRF) validation of RECIST 1.1 tumor images.

11. Study Identifiers & Governance

Primary ID: D3615C00001 **Registry Number:** NCT04305496 **Posting ID:** 246906003961237395 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** CAPitello-291 **Official Regulatory Title:** A Phase III Double-blind Randomised Study Assessing the Efficacy and Safety of Capivasertib + Fulvestrant Versus Placebo + Fulvestrant as Treatment for Locally Advanced (Inoperable) or Metastatic Hormone Receptor Positive, Human Epidermal Growth Factor Receptor 2 Negative (HR+/HER2-) Breast Cancer Following Recurrence or Progression On or After Treatment With an Aromatase Inhibitor

12. Clinical Framework (Oncology Specific)

Disease Area: Locally Advanced (Inoperable) or Metastatic Breast Cancer **Preferred UMLS Condition:** Metastatic estrogen receptor positive HER2 negative breast cancer **Diagnosis Threshold:** Recurrence or progression on/after aromatase inhibitor therapy **Medical Methodology:** Targeted AKT inhibition (Capivasertib) + Estrogen receptor antagonist (Fulvestrant) **Optimization Target:** Maximizing progression-free survival (PFS)

13. Study Procedures & Methodology

Management Pathway: Standard oncology guidelines for advanced HR+/HER2- breast cancer **Education Protocol (HCP):** Site initiation, protocol compliance training, and AE management (especially hyperglycemia and rash) **Education Protocol (Patient):** Careful counseling on the intermittent oral dosing schedule (4 days on, 3 days off per week) and recognition of dose-limiting

toxicities **Quality Audit Mechanism:** Centralized radiological review and structured PRO-CTCAE patient-reported outcome tracking

14. Observation & Follow-Up Schedule

Total Duration: Median follow-up ~13.0 months (until disease progression or unacceptable toxicity) **Onsite Visit Cadence:** Day 1 of every 28-day cycle (and Day 15 of Cycle 1) **Remote Monitoring Frequency:** Continuous monitoring per standard clinical protocol **Checklist Review Interval:** Regular scheduled tumor imaging (typically every 8 weeks) to evaluate RECIST 1.1 criteria

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				